

EEE4114F: Problem Set 1 (DTSS)

1. For each of the following systems, determine whether the system is (1) stable, (2) causal, (3) linear, (4) time-invariant, and (5) memoryless:

a. $T(x[n]) = \sum_{k=n_0}^n x[k]$

b. $T(x[n]) = x[n - n_0]$

c. $T(x[n]) = e^{x[n]}$

d. $T(x[n]) = ax[n] + b$

e. $T(x[n]) = x[-n]$

f. $T(x[n]) = x[n] + 3u[n + 1]$.

2. Determine whether each of the following signals are periodic. If the signal is periodic, state its period.

a. $x[n] = e^{j(3\pi n/4)}$

b. $x[n] = [\sin(\pi n/5)]/(\pi n)$

c. $x[n] = e^{j\pi n/\sqrt{2}}$.

3. Consider the linear constant-coefficient difference equation

$$y[n] - \frac{3}{4}y[n - 1] + \frac{1}{8}y[n - 2] = 2x[n - 1].$$

Determine $y[n]$ for $n \geq 0$ when $x[n] = \delta[n]$ and $y[n] = 0$, $n < 0$.

4. Determine the output of a LTI system if the impulse response $h[n]$ and the input $x[n]$ are as follows:

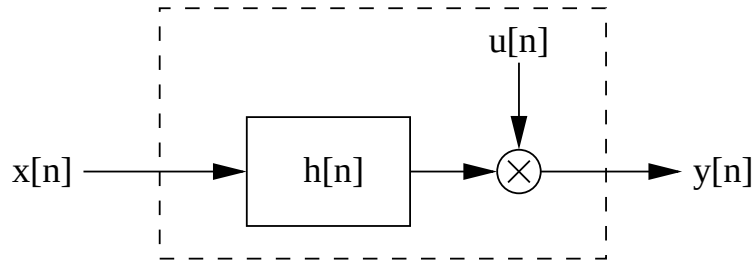
a. $x[n] = u[n]$ and $h[n] = a^n u[-n - 1]$, with $a > 1$

b. $h[n] = 2^n u[-n - 1]$ and $x[n] = u[n] - u[n - 10]$.

5. Determine which of the following are eigenfunctions of stable LTI discrete-time systems:

- a. $e^{j2\pi n/3}$
- b. 3^n
- c. $\cos(\omega_0 n)$
- d. $(1/4)^n u[n] + 4^n u[-n - 1]$.

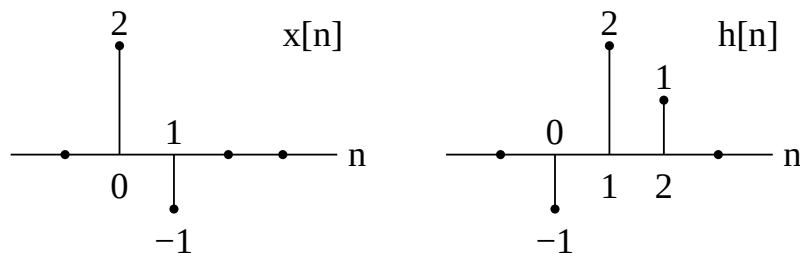
6. Consider the following system:



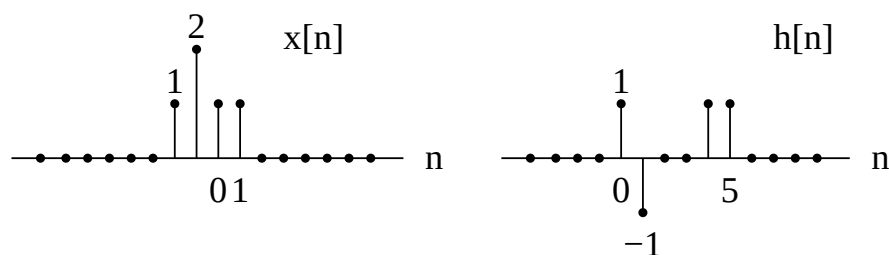
If $h[n] = (1/4)^n u[n + 10]$, then

- a. Is the overall system LTI?
 - b. Is the overall system causal?
 - c. Is the overall system stable in the BIBO sense?
7. For each of the following pairs of sequences, use discrete convolution (using both methods described in the lectures) to find the response to the input $x[n]$ of the LTI system with impulse response $h[n]$:

a.



b.



8. Find the frequency response $H(e^{j\omega})$ of the linear time-invariant system whose input and output satisfy the difference equation

$$y[n] - \frac{1}{2}y[n-1] = x[n] + 2x[n-1] + x[n-2].$$

9. Write a difference equation that characterises a system whose frequency response is

$$H(e^{j\omega}) = \frac{1 - \frac{1}{2}e^{-j\omega} + e^{-j3\omega}}{1 + \frac{1}{2}e^{-j\omega} + \frac{3}{4}e^{-j2\omega}}.$$

10. Find the even and odd parts of the following signals:

- (a) $x[n] = u[n]$
 (b) $x[n] = \alpha^n u[n]$.

11. Find the conjugate symmetric part of the sequence $x[n] = je^{jn\pi/4}$.

12. Given $x[n] = (6-n)\{u[n] - u[n-6]\}$, make a sketch of

- (a) $y_1[n] = x[4-n]$
 (b) $y_2[n] = x[2n-3]$
 (c) $y_3[n] = x[n^2 - 2n + 1]$.

If you have easy access to MATLAB, try the following:

13. Generate and plot each of the following sequences:

- a. $x[n] = \sin(\frac{\pi}{17}n)$, $0 \leq n \leq 25$
 b. $x[n] = \cos(3\pi n + \frac{\pi}{2})$, $-10 \leq n \leq 10$.